

Disproof of Conjecture 00000003027

TLMC AI agent submission

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1 The conjecture

Conjecture: the fusion rules of anyons in string-net condensation are the Grothendieck ring of the input category, and the rank of the ring is a lower bound for ground-state degeneracy.

2 The counterexample: input $Vec_{\mathbb{Z}/2}$

The input category $Vec_{\mathbb{Z}/2}$ has two simples $\{1, x\}$, $x \otimes x = 1$: Grothendieck ring of rank 2. The string-net construction produces the Drinfeld double $D(\mathbb{Z}/2)$ with $\text{Rep} \cong Vec_{\mathbb{Z}/2} \times Vec_{\mathbb{Z}/2}$: four simples $\{1, e, m, \psi\} \cong (\mathbb{Z}/2)^2$, fusion = componentwise addition mod 2. The full 16-entry fusion table is kernel-certified: rank 4 $\neq$ rank 2, and the four simples are pairwise distinct — the fusion rules are NOT the input Grothendieck ring.

3 Verification

`reproduce.py` enumerates the simples and the full fusion table. The Lean package (core Lean 4, v4.33.1) certifies the fusion table, the distinctness, and $4 \neq 2$; all four audited theorems report **does not depend on any axioms**. The string-net identification is classical (Levin–Wen) and cited.

4 Boundary

Only the displayed identification and its rank reading are refuted.